

Dissertation

A Distributed Edge FLISR Solution & Network Simulation Test Platform

Darren Leniston

Supervisor - Dr Kieran Murphy

Interim Report - May 2023



MSc in Computing (Enterprise Software Systems)
School Of Science and Computing

Abstract

The energy sector is facing wide-scale changes with the rapid introduction of disruptive distributed energy sources, renewables, electric vehicles and the changing relationship between the consumer and the utility. This has resulted in the need to leverage the capability of modern ICT and IoT technologies to meet the challenges presented by such changes. Of key concern is the resilience of the energy grid and the growing dependency on a stable supply to support business as usual activities and daily life for society at large. This details an attempt to realise a novel implementation of the concept of Fault Identification, Isolation and Service Restoration in a distributed fashion, enabled by Edge Computing techniques, to mitigate the effects of service loss to the end consumer and utility. The research focuses on the Irish energy grid in particular, exploring the background and motivation for such resiliency solutions, the related literature, research activities and the insights obtained from the gathered results.

Declaration

I have read and understood the Departmental policy on plagiarism.

I declare that this document is my own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education.

Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

Signature _____
Darren Leniston

Date _____

Contents

1	Introduction	1
1.1	Research background	1
1.2	Motivations for the research	2
1.3	Research objectives	2
2	Review of the literature	4
2.1	Digitalisation of the grid	4
2.2	Fault Location, Isolation & Service Restoration	4
2.3	Edge computing	6
2.4	Summary	6
3	Methods	8
3.1	Data & Setup of Experiment Environment	8
3.1.1	Network simulation platform	10
3.1.2	Trial site features	12
3.1.3	FLISR Edge solution & algorithm	12
3.2	Experiments	12
3.2.1	Network simulation test platform	12
3.2.2	Fault simulation operation	12
4	Results of the research	12
4.1	Research findings	12
4.2	Relevance to the DSO	12
4.3	Future work	12
5	Conclusion	13
6	Exploitation Activities	13
A	Appendices	16
A.0.1	FLISR Edge simulation platform user stories	16
A.0.2	FLISR Edge algorithm user stories	17

List of Figures

1	CML figures 2016-2020	2
2	Topology data ingested in graph database	8
3	Synergy data ERD	9
4	System stakeholders	10
5	FLISR simulation platform architecture	11

List of Tables

Glossary

Advanced Distribution Management System (ADMS) A software platform containing a suite of functions to support distribution network management and optimisation.

Customer Minutes Lost (CML) Measurement of the duration of interruption of supply over the course of a year. Consists of the average CML per consumer per year, in the event a supply interruption lasts up to or more than 3 minutes.

Distribution System Operator (DSO) Designated entity responsible for management of the distribution network from source of generation to final consumer.

Fault Passage Indicator (FPI) Indicator notifying that an on-pole switch has detected a fault on its specific point of the network.

Geographical Information System (GIS) Software system utilised to analyse and visualise data containing geographic information.

Load Break Fault Make Switch (LBFM) On-pole circuit breaker which can be controlled to energise or de-energise a section of network in the event of over-load, fault event or required maintenance.

Loss of Voltage Indicator (LVI) Indicator notifying that an on-pole switch has lost supply on its specific point of the network.

Low Voltage (LV) Network Distribution network which transmits to end customers, generally at mains rated voltage (230V single phase & 400V 3 phase for the Irish LV network).

Medium Voltage (MV) Network Distribution network which transmits at voltages above 1kV (kilovolt) (10kV, 20kV & 38kV for the Irish MV network).

List of Acronyms

CI Customer Interruption

DA Distribution Automation

FLISR Fault Location, Isolation and Service Restoration

ICT Information and Communication Technology

IGSD Intelligent Grid Switching Devices

IoT Internet of Things

KPI Key Performance Indicator

MAS Multi Agent Approach

1 Introduction

With the rapid changes as a result of the integration of advanced ICT & IoT technologies into business as usual and daily activities, this has led to large bodies of research in an effort to maximise the potential these technologies and their insights can unlock. Likewise, the energy sector has seen rapid changes with the introduction of renewable energy sources, electric vehicles and a drive to utilise software based solutions which leverage embedded devices. This drive towards the smart grid forms the backdrop of the following dissertation, in particular, this research proposal focuses on the aspect of energy grid resiliency. This dissertation expands on the learning's from previous related research as part of a European Horizon 2020 project [13], to develop a novel approach to an automated grid resilience strategy in the context of the Irish energy grid.

1.1 Research background

The energy grid is of vital importance in today's society and is a key piece of infrastructure which enables activities across a range of industries, services and domestic settings. Like many aspects of the modern world then, the energy industry has been seeing a paradigm shift in the last number of years with the introduction of state-of-the-art ICT and IoT technologies in an effort to modernise the energy sector; in terms of production, transmission and delivery to consumers [7]. These new technologies are taking the form of smart sensors and power monitoring at both the grid and domestic level, unlocking new data and granting greater insight into the state of the grid [3]. This, combined with controllable devices is leading to Advanced Distribution Management Systems (ADMS), enabling a fine-grained view and greater control of the energy grid.

There are a number of factors driving this race to modernise the grid. While traditional energy infrastructures operated on a top-down model, where supply to consumers from the transmission system follows a uni-directional flow and power generation is more centralised [14], this model of energy generation and supply is transforming at a rapid pace. With the introduction of distributed and renewable sources of energy production, electric vehicles, microgrids and the changing relationship between the consumer and the utility, whereby certain consumers are becoming "prosumers" (i.e., generating and selling their excess generation back to the grid) the structure of the energy grid is evolving [8]. In this way, the modernisation of the grid through ICT and IoT technologies intends to meet the challenges posed by the disruptive nature of these changes.

Within this context, the research pursued in this document aims to meet the requirement for a resilient energy grid, utilising modern technologies and techniques, to reduce and mitigate the effects of disruptive events on the grid and their impact on the consumer and utility. To meet this goal, the research leverages the concept of Fault Location, Isolation and Service Restoration (FLISR), a grid resilience strategy which in the event of a fault on the grid, aims to automate the identification of said fault, isolating damaged sections of the network and restoring service to consumers on lines effected by loss of service but are otherwise healthy and undamaged.

In combination with the FLISR technique, the research also explored the application of Edge Computing and novel representations of grid topology data to realise a distributed, fault-tolerant FLISR solution which is not fully reliant on a centralised cloud deployment, by moving subsets of the grid topology and FLISR logic to low-cost edge devices installed on existing grid infrastructure. In order to test, evaluate and demonstrate the solution, a simulation platform was also developed. This research is framed in the context of the Irish distribution grid, in particular three distribution grids of varying topology and features located throughout the country.

Note: Background, motivation and research objective sections content duplicated from proposal with minor edits

1.2 Motivations for the research

In tandem with the rapid transformation of the energy sector, the need for a resilient grid is vital with the growing reliance on consistent power delivery for societal, business and manufacturing activities. Coupled with this is the rising concern of climate change and its impact on everyday life. Recent times have seen an increase in the number and severity of weather events which cause localised and wide-scale damage to infrastructure. Such events often lead to loss of energy supply to consumers, businesses and other institutions through damage caused to wires, transformers, substations, equipment and other grid assets [18]. In relation to the Irish energy network, weather events such as ex-hurricane Ophelia in 2017 resulted in 5,500 damaged overhead lines and 385,000 homes & businesses without power [9].

In addition, the Distribution System Operator (DSO) in Ireland, ESB Networks, is faced with fines by the regulator based on specific KPIs which are directly influenced by the amount of unplanned service interruptions and loss of supply, namely Customer Interruption (CI) & Customer Minutes Lost (CML). For reference, Ophelia resulted in a total of 665.6 million CML, culminating in a 4.86 million euro fine imposed by the Irish energy regulator [9]. However, it must be said that although Ophelia was a rare event, 2017 was not an outlier in terms of total CML per customer, as detailed in ESB Networks 2020 report [16] and illustrated in Figure 1.

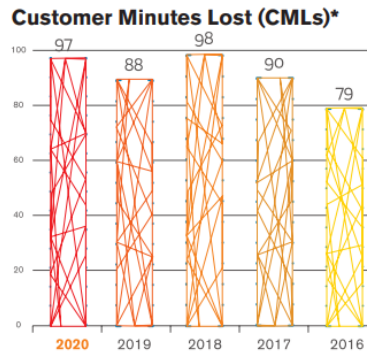


Figure 1: CML figures 2016-2020

In this way then, techniques such as FLISR offer the possibility to drastically mitigate against the effects of inclement weather events. However, there are limitations to the DSO in regards to deploying a FLISR solution to the grid, such as the cost of specialized equipment, installation and requirement for large-scale software platforms such as ADMS. In combination with this, weather events can also effect communication networks, thereby reducing or even negating the benefit of a cloud-based or centralised FLISR implementation. With this factor in mind, the concept of edge computing is one that may greatly benefit the operation of a FLISR algorithm in the field, by moving a subset of the business logic and data processing away from a centralised system, to the edge of the network. In the context of FLISR, this would enable sections of the network to operate independently if needed to isolate and restore fault events, providing a fault-tolerant FLISR solution able to operate effectively in conditions such as Ophelia.

1.3 Research objectives

The key objectives which framed the direction of the research activities centered around delivering a novel FLISR solution, leveraging modern technologies which can operate both centrally and in a distributed fashion at the network edge, and to test, evaluate and demonstrate the operation and effectiveness of the solution through the use of a simulation platform, utilising a number of grid topologies to assess the FLISR algorithms effectiveness against

a variety of network configurations located throughout Ireland.

The main output of the research is gauging how the developed FLISR solution may reduce CML in real-world operation, to that end, the following questions guided the research activities.

- **R1 - How effective will the outputs of the FLISR solution be in terms of reducing DSO fault KPIs?**

This question is crucial in regards to evaluating the value of the research activities and FLISR solution implemented, both compared to alternative approaches to the FLISR concept undertaken in other research, and in relation to the tested grid topologies and extrapolated to the Irish energy grid as a whole.

- **R2 - Given a similar set of conditions, how will the results produced by the centralised FLISR algorithm compare to the distributed FLISR algorithms operating locally at the edge of the network?**

A key feature of the distributed FLISR solution is to be in itself fault-tolerant, and therefore able to operate and produce results locally in the event communications with the central, “master” FLISR algorithm is unavailable. Given the limited view of the network granted to the FLISR algorithms operating locally at the edge, in comparison to the overall system awareness of the central FLISR algorithm, will the results produced by both be similar and, crucially, valid.

- **R3 - Given a diverse set of network topologies, will the FLISR solution output valid results for a variety of grid configurations?**

For the FLISR solution to be effective, a key requirement will be to operate and produce valid results given a wide set of grid topologies with varying network configurations in terms of structure, number of nodes and device types.

2 Review of the literature

The following section will explore and review a number of items of literature related to the research topics. The findings of this review are collated by a number of sub-themes; digitalisation of the grid, FLISR and edge computing, which directly relate to the concepts and technologies applied to the distributed edge FLISR solution.

2.1 Digitalisation of the grid

There is a wide range of literature available on the concept of the smart grid, and what this digitalisation and modernisation of the energy network means for the energy industry. *Fang et al.* [7] describe a smart grid as utilising two-way flows of both electricity and data in an aim to create an automated and distributed advanced energy delivery network, and through the use of modern ICT lead to a more efficient and flexible grid which can respond to a variety of events. The article in question describes the requirements for the smart grid from the perspective of three major systems; *smart infrastructure systems*, *smart management systems* and *smart protection systems*. Of most relevance here, are smart infrastructure systems, which are described as supporting two-way data flows via wired and wireless communication methods, and smart protection systems of which failure diagnosis and grid self-healing are a key factor.

An article by *Güngör et al.* [11] discuss the trends resulting in the need for a new, smarter energy infrastructure, taking the US as an example, reporting demand and consumption of electricity has increased by 2.5% annually over 20 years. This growing demand, in combination with factors such as greenhouse emissions, the environmental impact of extracting fossil-fuels for energy production and their growing scarcity, is driving the need for renewable sources and more flexibility in how the energy grid operates by enabling prosumer activities, microgrids, virtual power plants and increased grid resiliency to meet these challenges.

In the paper, the smart grid is described as a “modern electric power grid infrastructure for improved efficiency, reliability and safety, with smooth integration of renewable and alternative energy sources, through automated control and modern communication technologies”. In addition, they describe how the reliability of the energy infrastructure and supply can be supported and maintained in the smart grid, and how the impact of equipment failure, capacity constraints and natural accidents which lead to supply interruption and outages, can be mitigated by online power system monitoring, diagnostics and protection.

As described in the referenced articles, enabling the smart grid requires reliable information and data flow from a variety of smart devices both in the field and in the domestic setting, therefore the communication methods utilised are of key concern. In a separate article, *Güngör et al.* [10] discuss how wireless communication methods can meet some of the challenges presented by the move to the smart grid. In this paper, it is explained that while traditionally power system monitoring was realised through wired communication via cabling from pole to pole, this method is expensive and therefore not widely implemented today due to its financial impact.

Priya & Malhotra [19] describe how the need for efficient, reliable communication on the smart grid can be achieved by leveraging the capabilities of the telecommunication network, specifically referencing 5G and its fast, low-latency features and how these can meet the challenge of capturing large amounts of data from the field and enable quick, automated control services to manage, balance and ensure the resilience of the grid.

2.2 Fault Location, Isolation & Service Restoration

With resiliency of the network a key concern to the DSO and frequently mentioned as a requirement for the smart grid in the literature, FLISR is one such technique which has been explored and discussed in various research.

Note: Literature review section content duplicated from proposal with minor edits

Agüero [2] describes FLISR as a self-healing technique, though does make mention of certain authors preferring the term “self-restoration”, as in reality no system can currently fix physical damage to the grid without direct intervention by DSO personnel. The paper discusses how Distribution Automation (DA) technologies can enable remote monitoring, coordination and operation of distributed network assets at substation, feeder and customer level. Most relevant, is the description of utilising remote, on-pole switching devices as a method of sectionalising and restoring supply to customers in the event of a fault. The paper describes how the need for manual investigation and operating of switching devices on the part of DSO operation staff can be reduced or rendered redundant through automated FLISR, and the benefits of such a solution being the improvement of reliability statistics and the savings made due to the reduced cost of outages.

However, the text posits that leveraging such devices for self-healing is more appropriate for urban and suburban areas, rather than remote rural locations, where radial distribution is more commonplace (i.e., one source of power for consumers on the network). Though it could be argued that resilience strategies such as FLISR are more necessary for rural areas, due to the increased prevalence of faults as a result of deterioration and damage in remote locations.

A report produced by the US Department of Energy in 2014 [17] aimed to gather metrics on five separate FLISR solutions implemented throughout the United States over the course of one year. The findings included a large reduction in the number of customers affected by service loss (270,000) with 38 million fewer CML in total, translating to a reduction of 45% in the amount of interruptions and 51% in CML per fault event. The report specifies that an essential component for successful FLISR operations is the communications networks used to monitor and enable control over switching systems and devices, in combination with suitable interfaces which allow interoperability between various DSO systems and device types on the grid.

Each FLISR system explored utilises various device types, communication methods, modes of operation and location of the deployed system as part of its implementation. For example, a utility named CenterPoint trialed their FLISR system, named “Self-Healing Grid” which is supported by Intelligent Grid Switching Devices (IGSD) and requires manual validation by a System Operator in order to enact control operations. Conversely, a utility named Duke leveraged electronic reclosers, circuit breakers and line sensors for its FLISR system, dubbed “Self-Healing Teams” to enable a fully-automated solution.

The report specifies a number of learning’s gathered by the utilities from the FLISR trials, such as the need for the telecommunications network to be more resilient than the power delivery systems which rely upon them, and the necessity for the utility to consider the additional steps required to successfully deploy and operate a FLISR system. Steps identified included the testing and installation of new equipment, and the frequent firmware and software updates necessary to ensure the system continues to operate effectively due to its autonomous nature, though one utility, Georgia Power, found that developing a simulator to test and validate FLISR eliminated the need for extensive field trials and served as a training tool for operation staff.

A paper by *Eriksson et al.* [6] further discuss the limitations and considerations when implementing FLISR solutions, specifically centralised deployments, such as having a single point of failure and the requirement for the handling of large amounts of real-time data, large network topologies and the processing necessary to ensure consistent operation. They propose a distributed FLISR utilising a Multi Agent Approach (MAS) devices, tested against a radial network commonly found in remote areas, and applying Prim’s Algorithm to devise the FLISR logic. The solution proposed was tested against a cyber-physical testbed, with the results finding that the MAS approach to FLISR provided efficient grid stabilisation and fault restoration.

Most relevant is a report produced by ESB Networks as an output of the H2020 project SOGNO [15]. This project

focused on delivering turn-key grid services for power quality monitoring, estimating network state, load prediction and forecasting as well as FLISR. The FLISR implementation described in the report leveraged communication from Load Break Fault Make (LBFM) on-pole circuit breakers in the field. In the event of a fault, these switches communicate a Fault Passage Indicator (FPI) and Loss of Voltage Indicator and in this way it is possible to identify a fault occurrence. ESB Networks notes that a number of their networks currently rely upon manual intervention on the part of control room operation staff to manually identify and resolve fault events when they occur. This need for human intervention greatly increases the time taken to restore supply to consumers, leading to higher CML per event. While the FLISR solution tested as part of the SOGNO project produced good results, ESB Networks noted that its reliance on a centralised instance and telecommunications to operate is a severe limitation.

2.3 Edge computing

Over the last number of years, cloud computing has provided a method for large-scale data collection and processing for multiple use-cases and industries, however this dynamic is changing with the proliferation of IoT.

Shi et al. [20] describe this as the post-cloud era, where embedded IoT will not only serve as a data source but also consume and act on this data at the edge. As more data sources come online, the level of processing required by cloud computing services to collect, parse and act on this data increases. This, coupled with network bandwidth constraints when considering the large amounts of data gathered from distributed sources means new solutions are necessary and the paper proposes the edge computing concept as a solution to these challenges. The literature defines edge computing as enabling technologies which allow computing activities to be performed at the edge of the network on behalf of cloud-based services, therefore working as a data consumer as well as a data producer. One such example is the smartphone, which can serve as both a producer of data for cloud-based services and as a processing device at the edge. An example use-case being health-related applications, where user metrics are collected and processed on the device, and resulting health data is passed and stored on a cloud-based service.

Deploying and managing edge resources is a key factor when considering edge computing solutions, *Taleb et al.* [21] describe the importance of orchestration frameworks to enable integration and management of edge devices in terms of resource allocation and service management in regards to software and firmware updates. The paper describes how utilising virtualisation, such as containerised software services, supports in packaging, delivering, deploying and updating software services operating on edge computing devices located in the field.

In relation to the smart grid, a paper by *Zhang et al.* [23] proposes that a number of challenges facing the smart grid, such as high computational cost and transmission delay can be met through the application of edge computing. Similarly, *Chen et al.* [5] discuss the benefits of edge computing in supporting smart grid operations, with specific reference to grid resiliency and self-healing, describing how the dependency on centralised and cloud-based systems may lead to less efficient operation of grid control and monitoring services. Furthermore, the proximity of edge computing devices to grid assets may reduce latency when intelligent decisions are made locally, and provide finer-grained location awareness and monitoring of the network for the DSO, this is particularly important at the MV and LV levels where visibility of the grid from a system perspective is greatly reduced.

2.4 Summary

The exploration of the literature in the related thematic areas, highlights the level of research undertaken to realise the smart grid and the challenges faced in doing so. In particular, resiliency of the grid is referenced frequently as a key goal, with FLISR providing an effective method of achieving this objective. In regards to FLISR, there are many applications in the research that utilise various methods of fault identification and resolution in terms of the algorithms applied, communication protocols, standards and methods of control, such as fully autonomous

FLISR as well as those that require human intervention. A key concern identified is the need to ensure such grid resiliency solutions are in themselves fault-tolerant and therefore not dependent on centralised systems or core telecommunication networks. In reviewing the research, distributed FLISR solutions appear to be a means of resolving this limitation.

In realising a distributed FLISR solution then, edge computing emerges as an ideal paradigm to implement this functionality. By removing the need for large data processing and decision making functionality from cloud-based systems and moving subsets of this logic to the edge of the network, grid resiliency solutions, such as distributed FLISR, should benefit from reduced latency and dependency on core systems in order to operate effectively. Indeed, the literature describes how edge computing may be a key component in realising the smart grid, due to the amount of data generated and computation required to effectively manage the energy distribution system. In combination with this, is the difficulty in effecting large scale changes on the core distribution grid due to expense and dealing with legacy components. Therefore, enabling more computation, data collection and control at the edge of the network may afford smart grid functionality, implementation of concepts such as the VPP, microgrids and prosumer activities.

3 Methods

The following section will describe the experiments conducted to realise and evaluate both the FLISR network simulation platform and the FLISR algorithms, both in distributed and centralised modes, detailing the set up of the simulation platform and FLISR algorithms and the specific tests conducted to answer the research questions outlined in the objectives.

3.1 Data & Setup of Experiment Environment

The underlying data enabling the research activities is derived from a Geographical Information System (GIS) named Synergy, the data contained in this system maps the distribution network through the use of geospatial data [12] in addition to metadata related to each individual asset, whether this entity is a pole, transformer, feeder, switch, substation or other asset type. The properties contained within this metadata depend on the asset type, for example which nodes (poles) are connected, their phases (RST) which denote whether they are part of the three phase (MV) or single phase (LV) network, which poles are equipped with Load Break Fault Make (LBFM) circuit breakers and which of these are denoted as “Normally Open” and therefore serve as the boundary between network lines.

Before the data could be analysed, the export from Synergy which takes the form of a “MDB” (Microsoft Access Database) file was converted and ingested into a Postgres database, from here it was possible to generate an Entity Relationship Diagram (ERD), presented in Figure 3, to obtain a clearer view of the assets included in the data and their associated properties. From the analysis of the ERD it is clear that there is a deeply coupled relationship between entities in this data, and one of the goals of the research is to represent this data in a novel format through leveraging the concept of Graph Databases, which focus on the connections between data and affords a more ontological view [22] of the relationship between entities. To achieve this aim, the Synergy data was ingested into a “neo4j” graph database in a containerised environment as illustrated in Figure 2.

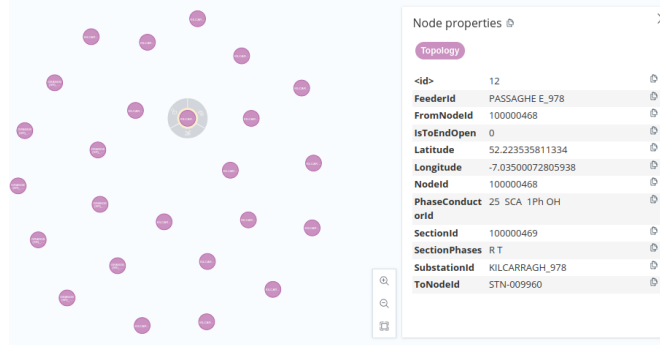


Figure 2: Topology data ingested in graph database



As described in the proposal document, the research work applies an iterative, agile model, in particular employing the concept of Behavior Driven Development (BDD) which aims to identify and define system functionality from the lens of key stakeholders and defines functions and business logic as “features” which guide the development process [1]. A system has one or more stakeholders, these are the actors outside of the system that interact with the application in question. Actors will have interests, goals or concerns in relation to that system, and may take the form of a person, outside system, subsystem or device to name a few. These actors will have various roles in regards to the operation of the system.

To define these actors, a list should be created with a basic format of:

- **Actor Name:** A clear identifier for the actor.
- **Actor Type:** One of person, device, system etc.
- **Actor Description:** Detailed description of the actor.
- **Actor Goal:** What the actor aims to achieve with the system.

It was necessary then to define users for both the simulation platform and the FLISR Edge algorithm itself, presented in Figure 4, and these served as the initial starting point to then define user stories which guided the development tasks. The individual user stories for each actor can be found in Appendix A.

FLISR Edge Simulation Platform Users

Actor Name	Actor Type	Actor Description	Actor Goal
DSO (Distribution System Operator)	Organisation (Utility)	Network Owner and Operator	They want a FLISR solution to reduce the level of CML caused by fault events on the distribution grid
CRO (Control Room Operator)	Person (DSO Personnel)	Asset Operator	They want to simulate fault events to capture FLISR Edge outputs to evaluate their effectiveness in the field

FLISR Edge Algorithm Users

Actor Name	Actor Type	Actor Description	Actor Goal
FLISR Master	System	FLISR Centralised Algorithm & Distributed System Orchestrator	Overall FLISR Edge system management, provisioning distributed algorithms to network edge, centralised algorithm with full network view
FLISR Node	Subsystem	Distributed FLISR algorithm	Subsystem which runs at network edge on individual switch, acts independently if connection to central system in unavailable

Figure 4: System stakeholders

3.1.1 Network simulation platform

The FLISR Edge simulation platform enables the simulation of fault events which will serve to both test and evaluate the FLISR algorithms in lieu of actual hardware, these simulation are based on the distribution grid topologies obtained from the Synergy GIS output described previously. The simulation platform leverages a micro service-like architecture, whereby services are loosely coupled, in addition these services are hosted within a containerised environment via “Docker”, and utilise “docker-compose” to enable simple deployment on either local systems or to a remote host [4]. Figure 5 describes the architecture implemented for the simulation platform with a description of each service following.

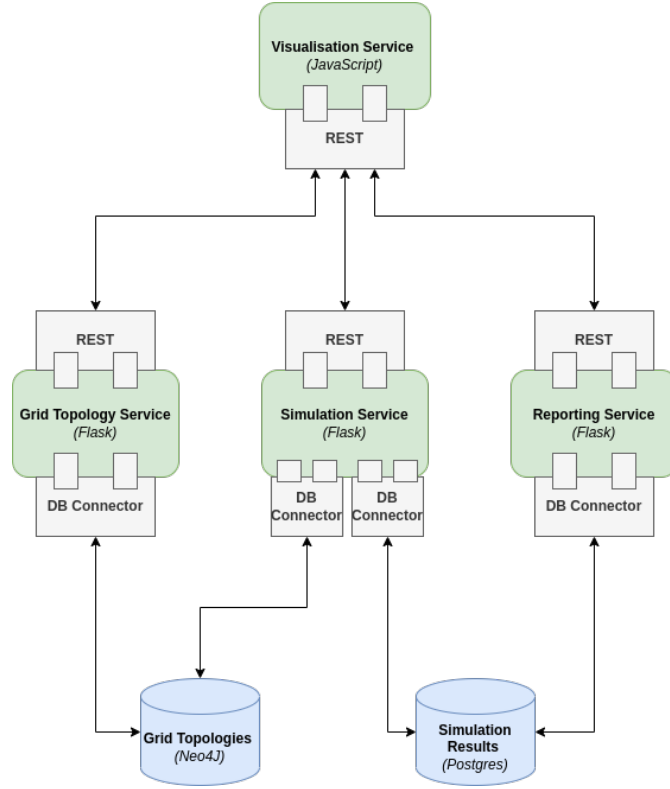


Figure 5: FLISR simulation platform architecture

- **Databases**

Two database services are in place which serve varying needs of the simulation platform, the first hosts the grid topologies with a “Neo4j” instance for both visualisation and simulation purposes. The second database is intended to host the simulation results in a relational SQL format, which can then be served to the user via browser or exported to CSV for analysis.

- **Grid topology service**

The grid topology service consists of a Python Flask server which provides an interface to request information relating to the grid topology, such as specific assets, asset data etc., In addition, this service generates “GeoJSON” from the grid topology data which is used to visualise the trial site distribution grids.

- **Simulation service**

The simulation service serves as the largest and most complex, which also consists of a Python Flask server. Unlike the grid topology service, the simulation service queries for specific data and contains functionality in order to run fault simulations and provide an outward interface to pass data to visualise these simulations. This service also interacts directly with the FLISR Edge algorithms and simulates control on the edge devices. On completion of a simulation, the results are stored in the relational database.

- **Reporting service**

The reporting service again consists of a Python Flask server and REST interface. The key purpose of this service is to query, parse and process simulation result data stored in the Postgres database, for visualisation to the user and for analysis purposes by exporting to CSV format.

- **Visualisation service**

Finally, the visualisation service consists of a React front-end which serves a graphical user interface to the user where the distribution grids and simulations are visualised. The UI contains controls to set up and run simulations, view the simulation process, simulation results and the option to export these results for analysis. “GeoJSON” and “LeafletJS” are leveraged to parse geolocation data, via “OpenStreetMap”.

3.1.2 Trial site features

Describe the trial site distribution grids, number of switches, make up etc., include screenshots from sim platform currently.

3.1.3 FLISR Edge solution & algorithm

Describe the FLISR Edge solution, how it is structured, orchestrated, the edge devices, algorithms etc.

3.2 Experiments

This section will describe in detail the actual experiments undertaken with the simulation platform and FLISR algorithms.

3.2.1 Network simulation test platform

Describe simulation platform and its structure, purpose, how it operates etc.

3.2.2 Fault simulation operation

Describe how the FLISR Edge solution and simulation platform interact and operate, how it generates results and how these are presented.

4 Results of the research

This section will describe the results gathered from the testing, including the findings, relevance of the results to the DSO and consumer, and future work to expand the solution further.

4.1 Research findings

This section will describe the findings as gleaned from the results gathered through the experiments, and describe these findings, how effective the solution is (or not), compare and contrast across the selected trial grids, centralised vs. distributed FLISR algorithm and any limitations.

4.2 Relevance to the DSO

This section will look to describe the relevance of the results to both the DSO and the consumer from a grid stability, financial and climate perspective in relation to weather events.

4.3 Future work

This section will describe future work which could be pursued to improve, expand or resolve any issues with the solution.

5 Conclusion

This section will summarise the document and outline the conclusions gathered as a whole from the research work, results, and relevance to the energy sector.

6 Exploitation Activities

Work has been undertaken to explore potential exploitation activities to maximise the outputs of the research, in an effort to gauge the interest in the research topic in the wider community and to gather insights and feedback from third parties and industry experts.

The first exploitation activity is the SETU Computing Department Open Day, which takes place on May 26th, the research topic was submitted for presentation and a poster has been developed to support this submission. The event will see participants from a variety of backgrounds, both academic and industry and will provide a useful backdrop to potentially gather useful feedback in regards to interest in the research topic from those that likely do not have a background in energy.

The second exploitation activity confirmed concerns the Association of Energy Engineers (AEE). AEE is a non-profit professional society with thousands of members across the globe who are actively engaged in the energy sector in both an industrial and scientific context. AEE was originally founded in the US, however the Irish chapter have set a conference to take place in the [SETU Arena on the 14th of September 2023](#). This research was submitted as an abstract and accepted, and will form one of the keynote presentations of the conference. This will present an invaluable opportunity to demonstrate the final research outputs to members of the energy sector and gather expert feedback to take into consideration.

In addition, the research will be leveraged and expanded to meet a use case for two potential upcoming Horizon Europe proposals in 2024, specifically the call “Resilient Infrastructure 2024 (HORIZON-CL3-2024-INFRA-01)” under the topic [Advanced real-time data analysis used for infrastructure resilience](#) and the call “Cross-sectoral solutions for the climate transition (HORIZON-CL5-2024-D2-01)” under the topic [Emerging energy technologies for a climate neutral Europe](#).

References

- [1] Hisham M. Abushama, Hanaa Altigani Alassam, and Fatin A. Elhaj. The effect of test-driven development and behavior-driven development on project success factors: A systematic literature review based study. In *2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE)*, pages 1–9, 2021.
- [2] Julio Romero Agüero. Applying self-healing schemes to modern power distribution systems. In *2012 IEEE Power and Energy Society General Meeting*, pages 1–4, 2012.
- [3] Ala Al-Fuqaha, Mohsen Guizani, Mehdi Mohammadi, Mohammed Aledhari, and Moussa Ayyash. Internet of things: A survey on enabling technologies, protocols, and applications. *IEEE Communications Surveys & Tutorials*, 17(4):2347–2376, 2015.
- [4] Armin Balalaie, Abbas Heydarnoori, and Pooyan Jamshidi. Microservices architecture enables devops: Migration to a cloud-native architecture. *IEEE Software*, 33(3):42–52, 2016.
- [5] Songlin Chen, Hong Wen, Jinsong Wu, Wenxin Lei, Wenjing Hou, Wenjie Liu, Aidong Xu, and Yixin Jiang. Internet of things based smart grids supported by intelligent edge computing. *IEEE Access*, 7:74089–74102, 2019.
- [6] Markus Eriksson, Mikel Armendariz, Oleg O. Vasilenko, Arshad Saleem, and Lars Nordström. Multiagent-based distribution automation solution for self-healing grids. *IEEE Transactions on Industrial Electronics*, 62(4):2620–2628, 2015.
- [7] Xi Fang, Satyajayant Misra, Guoliang Xue, and Dejun Yang. Smart grid — the new and improved power grid: A survey. *IEEE Communications Surveys & Tutorials*, 14(4):944–980, 2012.
- [8] Hassan Farhangi. The path of the smart grid. *IEEE Power and Energy Magazine*, 8(1):18–28, 2010.
- [9] Commission for Regulation of Utilities. 2017 performance report, 2018. <https://www.cru.ie/wp-content/uploads/2020/01/CRU20017-DSO-Annual-Performance-Report-APR-2017.pdf>.
- [10] Vehbi C. Gungor, Bin Lu, and Gerhard P. Hancke. Opportunities and challenges of wireless sensor networks in smart grid. *IEEE Transactions on Industrial Electronics*, 57(10):3557–3564, 2010.
- [11] Vehbi C. Gungor, Dilan Sahin, Taskin Kocak, Salih Ergut, Concettina Buccella, Carlo Cecati, and Gerhard P. Hancke. Smart grid technologies: Communication technologies and standards. *IEEE Transactions on Industrial Informatics*, 7(4):529–539, 2011.
- [12] Zhou Huang, Yu Fang, Bin Chen, and Xi Wu. Development of a grid gis prototype for geospatial data integration. In *2008 Seventh International Conference on Grid and Cooperative Computing*, pages 628–631, 2008.
- [13] D Leniston, D Ryan, C Power, P Hayes, and S Davy. Implementation of a software defined flisr solution on an active distribution grid [version 2; peer review: 2 approved]. *Open Research Europe*, 1(142), 2022.
- [14] Amir-Hamed Mohsenian-Rad, Vincent W. S. Wong, Juri Jatskevich, Robert Schober, and Alberto Leon-Garcia. Autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid. *IEEE Transactions on Smart Grid*, 1(3):320–331, 2010.

- [15] ESB Networks. Sogno h2020 project close-out report, 2020. https://www.esbnetworks.ie/docs/default-source/publications/sogno-h2020-project-close-out-report1d4eb2e4-f6ad-4a06-9c82-8c4510e2372f.pdf?sfvrsn=b1dc0ceb_11.
- [16] ESB Networks. Esb annual report 2020, 2021. https://esb.ie/docs/default-source/investor-relations-documents/esb-annual-financial-results-2020.pdf?sfvrsn=12f907f0_2.
- [17] U.S. Department of Energy. Fault location, isolation, and service restoration technologies reduce outage impact and duration, 2014. https://www.smartgrid.gov/files/documents/B5_draft_report-12-18-2014.pdf.
- [18] Mathaios Panteli and Pierluigi Mancarella. The grid: Stronger, bigger, smarter?: Presenting a conceptual framework of power system resilience. *IEEE Power and Energy Magazine*, 13(3):58–66, 2015.
- [19] Bhanu Priya and Jyoteesh Malhotra. An intelligent user-rat association for 5g enabled smart grid. In *2020 IEEE International Conference on Computing, Power and Communication Technologies (GUCON)*, pages 300–304, 2020.
- [20] Weisong Shi, Jie Cao, Quan Zhang, Youhuizi Li, and Lanyu Xu. Edge computing: Vision and challenges. *IEEE Internet of Things Journal*, 3(5):637–646, 2016.
- [21] Tarik Taleb, Konstantinos Samdanis, Badr Mada, Hannu Flinck, Sunny Dutta, and Dario Sabella. On multi-access edge computing: A survey of the emerging 5g network edge cloud architecture and orchestration. *IEEE Communications Surveys & Tutorials*, 19(3):1657–1681, 2017.
- [22] H.R. Vyawahare, P.P. Karde, and V.M. Thakare. A hybrid database approach using graph and relational database. In *2018 International Conference on Research in Intelligent and Computing in Engineering (RICE)*, pages 1–4, 2018.
- [23] Bin Zhang, Chong Wang, Yumin Chen, Jian Fu, and Yangjun Zhou. An edge data access control scheme suitable for wireless smart grid environment. In *2021 8th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud)/2021 7th IEEE International Conference on Edge Computing and Scalable Cloud (EdgeCom)*, pages 182–187, 2021.

A Appendices

A.0.1 FLISR Edge simulation platform user stories

CRO User Stories

User Story Number	As a "role"	I want to "goal/desire"	So that "benefit"
CRO-01	As a CRO	I want to view a distribution grid on a geographical map	I can see the structure of the distribution grid
CRO-02	As a CRO	I want to view LBFM switches deployed on a distribution grid	I can see the number of on pole circuit breakers on a distribution grid
CRO-03	As a CRO	I want to view LBFM metadata	I can read the specifications of selected switches
CRO-04	As a CRO	I want to view distribution grid substations	I can see the location of substations feeding the distribution grid
CRO-05	As a CRO	I want to visualise the flow of supply	I can see how energy supply is proliferating throughout the network
CRO-06	As a CRO	I want to simulate a fault on the grid	I can test the effects a specific fault event will have on the energy supply
CRO-07	As a CRO	I want a simulated fault to be resolved by a FLISR algorithm	I can see the effect the FLISR algorithm has in response to a simulated fault event

DSO User Stories

User Story Number	As a "role"	I want to "goal/desire"	So that "benefit"
DSO-01	As a DSO	I want access to customer metrics	I have context for the amount of MV and LV customers on a given distribution grid
DSO-02	As a DSO	I want to view the CML of a simulated fault event	I can report on the effect a fault event has on customers
DSO-03	As a DSO	I want to view an approximation of the cost of a simulated fault event	I can report on the financial penalty of a fault event

A.0.2 FLISR Edge algorithm user stories

FLISR Master User Stories

User Story Number	As a "role"	I want to "goal/desire"	So that "benefit"
FM-01	As a FLISR Master	I want to have a central view of the distribution grid	I can see the structure of the distribution grid
FM-02	As a FLISR Master	I want to have a central FLISR algorithm	I can resolve faults from a central location
FM-03	As a FLISR Master	I want a view of edge devices	I can monitor edge devices installed on the grid
FM-04	As a FLISR Master	I want to be able to provision a network topology subset	FLISR nodes have a subset of the network related to them available
FM-05	As a FLISR Master	I want to provision a distributed FLISR algorithm	FLISR nodes have a subset of the FLISR algorithm
FM-06	As a FLISR Master	I want to receive a heartbeat from FLISR nodes	I can monitor the connection of FLISR nodes on the grid
FM-07	As a FLISR Master	I want to receive the status of LBFM switches from FLISR Nodes	I know when a fault event occurs
FM-08	As a FLISR Master	I want to send commands to FLISR Nodes	I can action on fault restoration steps

FLISR Node User Stories

User Story Number	As a "role"	I want to "goal/desire"	So that "benefit"
FN-01	As a FLISR Node	I want to have a subset of the network topology	I can see the structure of the line I am located
FN-02	As a FLISR Node	I want to have a subset of the FLISR algorithm	I can resolve a fault event
FN-03	As a FLISR Node	I want to monitor the status of a switch	I can detect FPI and/or LVI
FN-04	As a FLISR Node	I want to send switch status to the FLISR Master	in the event of a fault the status of the switch is sent to the FLISR master to process
FN-05	As a FLISR Node	I want to send a heartbeat to the FLISR Master	I can determine if communication is available
FN-06	As a FLISR Node	I want to act independently if communication with the FLISR Master is not available	I can use the subset of the FLISR algorithm to operate a switch if communication is not possible